

CHM204 : Electrochemistry, colloids and interfaces

Mid-semester examination 2 AY 2025-26

Full marks 30 : Time 60 minutes

Answer all the questions

Elaborate on all the relevant steps for part marking

Mobile phone is not allowed for calculation

A scientific calculator should be used for any calculations

MS24056

DWYA JYOTI RAY

1. At 25°C, the specific conductance of a solution containing 3.20 g of BaCl<sub>2</sub> per litre is  $4.10 \times 10^{-3}$  mho cm<sup>-1</sup>. Calculate the equivalent conductance of the solution (Molar mass BaCl<sub>2</sub> = 208 g/mol) [4]
2. The equivalent conductance of a 0.014(N) Solution of ClCH<sub>2</sub>COOH, is 109.0 mho cm<sup>2</sup>. The ion conductance is 40.2 and 349.8 mho cm<sup>2</sup>, respectively. Calculate the degree of dissociation and the dissociation constant of ClCH<sub>2</sub>COOH. [4]
3. Calculate the ionic strength of a solution containing a mixture of 0.02 M AlCl<sub>3</sub> and 0.01 M KNO<sub>3</sub>. [4]
4. If the specific conductance of SrSO<sub>4</sub> in a saturated solution at 298K is  $1.5 \times 10^{-4}$  and that of pure water is  $1.5 \times 10^{-6}$  ohm<sup>-1</sup> cm<sup>-1</sup>, what is the solubility of SrSO<sub>4</sub> in water at 298K? [Given  $l_+ = 59.2$ ,  $l_- = 79.8$ , Eq. wt of SrSO<sub>4</sub> = 92.8]. [4]
5. Draw the conductometric titration plot for the titration of a triple mixture of sulphuric acid, ammonium sulphate, and potassium sulphate using suitable titrating agent(s) and explain the plot briefly. [8]
6. If at 18 °C, the mobilities of ammonium and chlorate ions are  $6.6 \times 10^{-4}$  and  $5.7 \times 10^{-4}$  cm<sup>2</sup>/volt/s, then what is the equivalent conductance of ammonium chlorate and the transport number of the cation? [6]

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Model Solution.

Sol Q1.

The normality of  $\text{BaCl}_2$  solution =  $\frac{\text{g/L}}{\text{eq. wt.}} = \frac{3.2 \text{ g/L}}{102 \text{ g/eq.}} = 3.077 \times 10^{-2} \text{ eqv. L}^{-1}$  (1)

$\therefore$  Equivalent conductance,  $\lambda = \frac{1000 L}{N}$  (2)

$$= \frac{1000 \times 4.10 \times 10^{-3} \text{ mho cm}^{-1}}{3.077 \times 10^{-2} \text{ eqv. L}^{-1}}$$

(1)

$$= 133.25 \text{ mho cm}^2 \text{ eqv}^{-1}$$

Sol Q2.

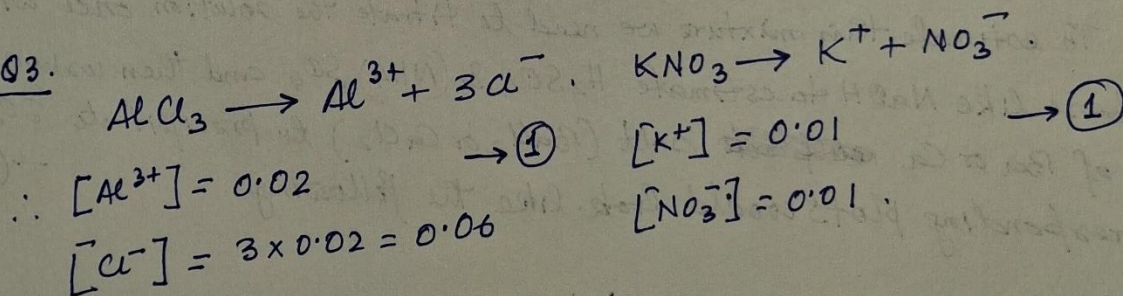
$\alpha = \frac{\Lambda_c}{\Lambda_0}$ ,  $\Lambda_0 = \text{sum of ion conductances}$

$\therefore \Lambda_0 = (40.2 + 349.8) = 390$  (2)

$\therefore \alpha = \frac{109}{390} = 0.279$

and,  $K_a = \frac{C\alpha^2}{(1-\alpha)} = \frac{0.014 \times (0.279)^2}{1-0.279} = 1.5 \times 10^{-3}$  (2)

Sol Q3.



ionic strength,  $I = \frac{1}{2} \sum C_i z_i^2$

$\therefore I = \frac{1}{2} [(0.02) \times 3^2 + (0.06) \times 1^2 + (0.01) \times 1^2 + (0.01) \times 1^2]$  (2)

$= 0.13$



Sol Q4.

Given,  $L_s = 1.5 \times 10^{-4} \text{ mho cm}^{-1}$   
 $L_w = 1.5 \times 10^{-6} \text{ mho cm}^{-1}$

$L_{\text{salt}} = L_s - L_w = 1.485 \times 10^{-4} \text{ mho cm}^{-1}$

$\Lambda_0 = l_+^0 + l_-^0$

$= (59.2 + 79.8) \text{ mho cm}^2 \text{ eqv}^{-1} = 139 \text{ mho cm}^2 \text{ eqv}^{-1}$

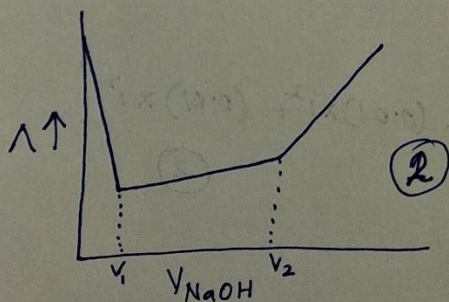
Again,  $\Lambda_0 = \frac{1000 \times L_s}{C}$

or  $C = \frac{1000 \times L_s}{\Lambda_0} = \frac{1000 \times 1.485 \times 10^{-4} \text{ mho cm}^{-1}}{139 \text{ mho cm}^2 \text{ eqv}^{-1}}$

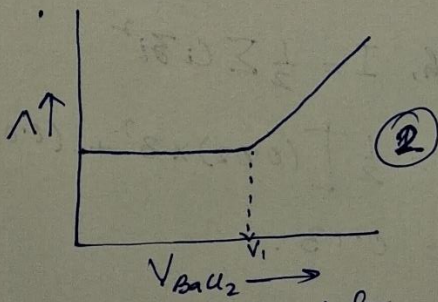
$= 0.00107 \text{ eqv. l}^{-1}$

$\therefore \text{Solubility} = \text{conc} \times \text{eqv. wt}$   
 $= 0.00107 \times 92.8 \text{ g l}^{-1}$   
 $= 0.099 \text{ g l}^{-1}$

Sol Q5. To estimate this mixture we need to titrate the solution once with a strong base like NaOH to estimate  $\text{H}_2\text{SO}_4$  &  $(\text{NH}_4)_2\text{SO}_4$  and then using a solution of Ba or Ca salt ( $\text{BaCl}_2$  or  $\text{CaCl}_2$ ) to precipitate  $\text{SO}_4^{2-}$ .  
 The corresponding plots would look like the following.



$V_1 = \text{vol of NaOH req. for } \text{H}_2\text{SO}_4$   
 $V_2 - V_1 = \text{ " " " " } (\text{NH}_4)_2\text{SO}_4$   
 conc. of  $\text{H}_2\text{SO}_4$  &  $(\text{NH}_4)_2\text{SO}_4$  need to be calc. in g/l unit.



$V_1 = \text{vol. of } \text{BaCl}_2 \text{ req. for total } \text{SO}_4^{2-}$   
 Need to calc.  $\text{SO}_4^{2-}$  in terms of g/l of  $\text{H}_2\text{SO}_4$  and hence amount of  $\text{K}_2\text{SO}_4$  need to be calc.

Sol Q6.

$$u_+ = 6.6 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$u_- = 5.7 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$$

$$\therefore \lambda_0 = F \times (u_+ + u_-) \quad \text{--- (1)}$$

$$= 96500 \times (6.6 \times 10^{-4} + 5.7 \times 10^{-4}) \text{ mho cm}^2 \text{ eqv}^{-1}$$

$$= 118.7 \text{ mho cm}^2 \text{ eqv}^{-1} \quad \text{--- (2)}$$

$$t_+ = \frac{u_+}{u_+ + u_-} \quad \text{--- (1)}$$

$$= 0.537 \quad \text{--- (2)}$$